A sharp polygonal Finsler–Hadwiger inequality

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Abstract

For every convex n-gon of positive area A with side lengths $a_1, \dots, a_n$, we prove

$$\sum_{i=1}^{n} a_i^2 \geq 4 \tan \frac{\pi}{n} A + c_n \sum_{i<j} (a_i - a_j)^2, \qquad c_n = 1 - \frac{\tan(\pi/n)}{\tan(\pi/(n-1))},$$

where $c_3 = 1$. The coefficient is optimal, and equality holds exactly for regular polygons. Sharpness is approached by cyclic polygons with one side tending to zero and the other sides tending to those of a regular $(n-1)$-gon, or a doubled segment when $n = 3$. The proof combines the classical maximal-area property of cyclic polygons with a two-variable minimization on the simplex of half central angles. Equivalently, we determine the best constant controlling the side-length variance by the polygonal isoperimetric deficit. Combined with a classical radial-variance inequality and the established convexification reduction, the estimate recovers the main stability estimates of Indrei and Nurbekyan with explicit constants. The cases $n = 3$ and $n = 4$ recover known Finsler–Hadwiger inequalities.

Keywords. Polygonal isoperimetric inequality; Finsler–Hadwiger inequality; side-length variance; cyclic polygon; sharp constant.

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1 Introduction

Let P be a convex n-gon with perimeter L, area A, and side lengths $a_1, \dots, a_n$. The polygonal isoperimetric inequality is

$$L^2 \geq 4n \tan \frac{\pi}{n} A, \tag{1.1}$$

with equality exactly for the regular n-gon; see, for example, [9]. By Cauchy–Schwarz, it implies

$$\sum_{i=1}^{n} a_i^2 \geq 4 \tan \frac{\pi}{n} A. \tag{1.2}$$

For triangles, the Finsler–Hadwiger inequality [3] provides the stronger estimate

$$a^2 + b^2 + c^2 \geq 4\sqrt{3}\, A + (a-b)^2 + (b-c)^2 + (c-a)^2.$$

For quadrilaterals, Giugiuc, Dao and Altintas [5, Theorem 1.1] proved the corresponding inequality with coefficient $1 - 1/\sqrt{3}$ in front of the sum of the six squared side differences.

The question of a refinement of this form for general n was posed explicitly by Dao [2]. The coefficient suggested there, $\tan(\pi/n) - \tan(\pi/(n+2))$, is too large already for $n = 3$,

as Pinelis observed in his answer to that question. In this note we determine the optimal coefficient for every $n \geq 3$.

The estimate belongs to quantitative polygonal isoperimetry. Indrei and Nurbekyan [7] proved that the isoperimetric deficit controls the variances of the side lengths and of the distances from the vertices to their barycenter. Indrei [6] subsequently included a term measuring angular variance. Those results control several geometric defects simultaneously; here we determine the best coefficient of the side-length term. Combined with the classical radial estimate and the same convexification reduction used in their proof, our bound recovers their Theorem 1.1 and Corollaries 1.2–1.3 with explicit constants; the derivation is given in Section 6. For triangles, optimal nonlinear relations between area, perimeter and side-length dispersion were obtained in [1].

Our proof is global and elementary after a classical geometric reduction. We first replace a polygon by the cyclic polygon with the same side lengths and maximal area. A two-variable minimization shows that the resulting trigonometric functional has a minimum at a point whose positive coordinates are equal. Concavity of $x \mapsto \cot(\pi/x)$ then singles out the configuration with $n-1$ positive coordinates. Thus the sharp coefficient is dictated by a degeneration, rather than by perturbations of a regular n-gon.

2 The sharp inequality and its quantitative forms

Throughout, a convex n-gon has distinct vertices in boundary order, positive side lengths and positive area. Consecutive collinear sides are allowed. Set

$$\tau_n = \tan\frac{\pi}{n} \quad (n \geq 3), \qquad \tau_2 = +\infty,$$

and interpret τ_n/τ_2 as zero.

Theorem 2.1. *For every convex n-gon P, $n \geq 3$, with side lengths $a_1, \dots, a_n$ and area A,*

$$\sum_{i=1}^{n} a_i^2 \geq 4\tau_n A + c_n \sum_{i<j} (a_i - a_j)^2, \qquad c_n = 1 - \frac{\tau_n}{\tau_{n-1}}. \tag{2.1}$$

The coefficient c_n is the largest possible. Equality holds if and only if P is regular.

Write

$$D(P) = \sum_{i<j} (a_i - a_j)^2, \qquad \delta_n(P) = L^2 - 4n\tau_n A,$$
$$V(P) = \sum_{i=1}^{n} \left(a_i - \frac{L}{n}\right)^2, \qquad \sigma_s^2(P) = \frac{1}{n} V(P).$$

The identities

$$D(P) = n\sum_i a_i^2 - L^2 = nV(P) \tag{2.2}$$

give the following equivalent formulation.

Corollary 2.2. *Under the assumptions of Theorem 2.1,*

$$\delta_n(P) \geq \gamma_n D(P), \qquad \gamma_n = nc_n - 1 = (n-1) - n\frac{\tau_n}{\tau_{n-1}} > 0. \tag{2.3}$$

Equivalently,

$$V(P) \le \frac{\delta_n(P)}{n\gamma_n}, \qquad \sigma_s^2(P) \le \frac{\delta_n(P)}{n^2\gamma_n}. \tag{2.4}$$

The coefficient γ_n is maximal and the two reciprocal coefficients in (2.4) *are minimal. Equality holds exactly for regular polygons.*

Indeed, subtracting the right-hand side of (2.1) from its left-hand side and multiplying by n gives $\delta_n(P) - (nc_n - 1)D(P)$, proving equivalence. For $n = 3$, $\gamma_3 = 2$. For $n \ge 4$, positivity follows because $x\tan(\pi/x)$ is strictly decreasing for $x > 2$: its derivative is $\tan u - u\sec^2 u < 0$, where $u = \pi/x \in (0, \pi/2)$.

3 Cyclic reduction and two-variable minimization

For fixed positive side lengths satisfying the strict polygon inequalities, a maximal-area convex polygon is cyclic, and equality in the area comparison forces cyclicity. This is the classical secant-polygon theorem; see [8, Section 1, Theorem B], which also discusses existence and uniqueness of the cyclic polygon. The side lengths of P satisfy the strict polygon inequalities because P has positive area. Since the coefficient of A in (2.1) is positive, it therefore suffices to prove the inequality for cyclic polygons.

Let R be the circumradius, and let $2x_i$ denote the counterclockwise angular gap from the ith vertex to the next. Then

$$0 < x_i < \pi, \qquad \sum_{i=1}^{n} x_i = \pi, \qquad a_i = 2R\sin x_i. \tag{3.1}$$

The circumcenter may lie outside the polygon; one of the x_i may exceed $\pi/2$. The signed triangle decomposition about the circumcenter nevertheless gives

$$A = \frac{R^2}{2}\sum_i \sin(2x_i) = R^2\sum_i \sin x_i \cos x_i. \tag{3.2}$$

Put

$$S = \sum_i \sin x_i, \qquad U = \sum_i \sin^2 x_i, \qquad B = \sum_i \sin x_i \cos x_i.$$

By (2.2), the difference between the two sides of (2.1), divided by $4R^2$, is

$$\Phi_c(x) = cS^2 - (nc - 1)U - \tau_n B \tag{3.3}$$

with $c = c_n$. We minimize this continuous function on

$$\Delta_n = \left\{ x \in [0,\pi]^n : \sum_i x_i = \pi \right\}.$$

Lemma 3.1 (Two-variable minimization)**.** *Fix $n \ge 3$ and $c > 0$. The minimum of Φ_c on Δ_n is attained, up to permutation, at*

$$x^{(m)} = \left(\underbrace{\pi/m, \ldots, \pi/m}_{m \text{ entries}}, 0, \ldots, 0 \right) \quad \text{for some } 1 \le m \le n. \tag{3.4}$$

Moreover, every minimizer with at least three positive coordinates has all its positive coordinates equal.

Proof. A minimum exists by compactness. Fix all but two coordinates x, y, and fix their sum $p = x + y$. If $p > 0$, write

$$x = \frac{p}{2} + d, \qquad y = \frac{p}{2} - d, \qquad z = \cos d.$$

Since $0 \le p \le \pi$, the feasible interval is $\cos(p/2) \le z \le 1$. Let r be the sum of the sines of the remaining coordinates. The pair contributions satisfy

$$\begin{aligned} \sin x + \sin y &= 2\sin(p/2)z, \\ \sin^2 x + \sin^2 y &= 1 - \cos p\,(2z^2 - 1), \\ \sin x \cos x + \sin y \cos y &= \sin p\,(2z^2 - 1). \end{aligned}$$

Substitution in (3.3) gives

$$\Phi_c = \alpha z^2 + \beta z + \eta, \tag{3.5}$$

where η is independent of z and

$$\begin{aligned} \alpha &= 4c\sin^2(p/2) + 2(nc - 1)\cos p - 2\tau_n \sin p, \\ \beta &= 4cr\sin(p/2) \ge 0. \end{aligned} \tag{3.6}$$

Suppose a global minimizer has at least three positive coordinates, and select two unequal positive ones x, y. Then $p < \pi$, $r > 0$, and $\beta > 0$. Moreover, z lies strictly between $\cos(p/2)$ and 1. If $\alpha \ge 0$, the derivative $2\alpha z + \beta$ is strictly positive, so decreasing z decreases Φ_c. If $\alpha < 0$, strict concavity implies that one endpoint has a strictly smaller value. Either case contradicts global minimality. Thus every positive pair in such a minimizer is equal.

If a minimizer has exactly two positive coordinates, then $p = \pi$ and $r = 0$. The quadratic (3.5) becomes $\alpha z^2 + \eta$ on $[0, 1]$, whose minimum can be chosen at an endpoint. These endpoints have respectively one positive coordinate or two equal positive coordinates. A minimizer with one positive coordinate already has the required form. Since the coordinates sum to π, this proves (3.4). □

Remark 3.2 (Two positive coordinates). On an edge of the simplex, direct substitution gives

$$\Phi_c(x, \pi - x, 0, \dots, 0) = 2\big(1 - (n-2)c\big)\sin^2 x.$$

If $1 - (n-2)c = 0$, this restriction is constant. In particular, for $n = 3$ and $c = c_3 = 1$, every point of such an edge has value zero, including points whose positive coordinates are unequal. Lemma 3.1 asserts only that a minimizer with equal positive coordinates can be chosen; its characterization of every minimizer requires at least three positive coordinates. These edge configurations have zero area and hence yield no additional equality cases in Theorem 2.1.

4 Proof of the optimal coefficient

At $x^{(1)}$, the functional is zero. This point represents no positive-area polygon. For $m \ge 2$, substitution of

$$S = m\sin\frac{\pi}{m}, \quad U = m\sin^2\frac{\pi}{m}, \quad B = m\sin\frac{\pi}{m}\cos\frac{\pi}{m}$$

into (3.3) yields

$$\Phi_c(x^{(m)}) = m\sin^2\frac{\pi}{m}\left(1 - c(n-m) - \tau_n \cot\frac{\pi}{m}\right). \tag{4.1}$$

The value for $m = n$ is zero for every c. For $2 \le m < n$, nonnegativity is equivalent to

$$c \le c_{n,m} := \frac{1 - \tau_n/\tau_m}{n - m}. \tag{4.2}$$

Lemma 4.1. *For $n \ge 3$ and $2 \le m < n$, $c_{n,m} \ge c_{n,n-1}$, with strict inequality if $m < n - 1$.*

Proof. Define $h(t) = \cot(\pi/t)$ for $t \ge 2$. With $u = \pi/t$,

$$h''(t) = -\frac{2u^3}{\pi^2 \sin^2 u}(1 - u\cot u) < 0 \qquad (t > 2), \tag{4.3}$$

because $\tan u > u$ for $0 < u < \pi/2$. Hence h is strictly concave on $[2, \infty)$. Since $\tau_n h(n) = 1$,

$$c_{n,m} = \tau_n \frac{h(n) - h(m)}{n - m} \ge \tau_n\big(h(n) - h(n-1)\big) = c_{n,n-1}.$$

Strict concavity gives strict inequality for $m < n - 1$. □

Proof of Theorem 2.1. Take $c = c_n = c_{n,n-1} > 0$. Lemma 4.1 and (4.1) show that $\Phi_{c_n}(x^{(m)}) \ge 0$ for every m. Lemma 3.1 therefore implies $\Phi_{c_n} \ge 0$ on Δ_n. The cyclic reduction proves (2.1) for every convex polygon under consideration.

Suppose equality holds. Equality in the area comparison first forces P to be cyclic. All coordinates in (3.1) are then positive. Since the global minimum of Φ_{c_n} is zero, the final assertion of Lemma 3.1 implies $x_1 = \cdots = x_n = \pi/n$. Thus P is regular. Conversely, direct substitution gives equality for regular polygons.

For sharpness, consider cyclic polygons of circumradius 1 with half central angles

$$x_1 = \cdots = x_{n-1} = \frac{\pi - \varepsilon}{n-1}, \qquad x_n = \varepsilon, \qquad 0 < \varepsilon < \frac{\pi}{n}. \tag{4.4}$$

They are strictly convex and tend to $x^{(n-1)}$ as $\varepsilon \downarrow 0$. For $c > c_n$, (4.1) gives

$$\Phi_c(x^{(n-1)}) = -(n-1)\sin^2\frac{\pi}{n-1}\,(c - c_n) < 0.$$

Continuity shows that the proposed inequality fails for sufficiently small positive ε. This proves optimality. □

For $n \ge 4$, the sequence (4.4) converges to a regular $(n-1)$-gon with one repeated vertex. For $n = 3$, it converges to a segment traversed in both directions, with limiting side lengths $(2, 2, 0)$. In either case the limiting side-length defect is $4(n-1)\sin^2(\pi/(n-1)) > 0$. Thus the optimal constants in Corollary 2.2 are also approached by this sequence.

5 Special cases and asymptotics

For $n = 3$, we have $c_3 = 1$ and $\gamma_3 = 2$, recovering the Finsler–Hadwiger inequality. For $n = 4$,

$$c_4 = 1 - \frac{1}{\sqrt{3}}, \qquad \gamma_4 = 3 - \frac{4}{\sqrt{3}}.$$

To compare directly with [5, Theorem 1.1], write $Q = a^2 + b^2 + c^2 + d^2$ and $T = ab + ac + ad + bc + bd + cd$. Since $D = 3Q - 2T$, (2.1) becomes

$$A \le \frac{T}{3 + \sqrt{3}} - \frac{Q}{2(\sqrt{3}+1)^2},$$

which is precisely their formulation.

As $n \to \infty$, expansion of the tangent gives

$$\gamma_n = \frac{2\pi^2}{3n^2} + O(n^{-3}), \qquad \frac{1}{n\gamma_n} \sim \frac{3n}{2\pi^2}, \qquad \frac{1}{n^2\gamma_n} \longrightarrow \frac{3}{2\pi^2}. \tag{5.1}$$

The normalization of the variance therefore matters when comparing constants for different numbers of sides.

6 Recovery of side and radial stability

Let $v_1, \dots, v_n$ be the vertices of P, let $g = n^{-1}\sum_i v_i$ be their barycenter, and set

$$r_i = |v_i - g|, \qquad \bar{r} = \frac{1}{n}\sum_i r_i, \qquad \sigma_r^2(P) = \frac{1}{n}\sum_i (r_i - \bar{r})^2.$$

These are distances to the vertex barycenter, not to a circumcenter. The classical radial-variance inequality of Fisher, Ruoff and Shilleto [4], in the form recorded in [7, Section 3.1, equations (3.1)–(3.2)], states

$$8n^2 \sin^2\frac{\pi}{n}\, \sigma_r^2(P) \le \delta_n(P) + n^2\sigma_s^2(P). \tag{6.1}$$

Thus the side-length estimate supplies the remaining ingredient in the following joint stability bound.

Corollary 6.1. *For every convex n-gon of positive area,*

$$\sigma_s^2(P) + \sigma_r^2(P) \le C_n \delta_n(P), \qquad C_n = \frac{1}{n^2\gamma_n} + \frac{1 + \gamma_n^{-1}}{8n^2\sin^2(\pi/n)}. \tag{6.2}$$

Proof. By (2.4), $n^2\sigma_s^2(P) \le \gamma_n^{-1}\delta_n(P)$. Substituting in (6.1) gives

$$\sigma_r^2(P) \le \frac{1 + \gamma_n^{-1}}{8n^2\sin^2(\pi/n)}\, \delta_n(P).$$

Adding the side-length bound proves (6.2). □

This recovers [7, Corollary 1.3] with an explicit constant. We do not assert that C_n is optimal for the sum of the two variances; the sharp coefficient established here is the one in the side-length bound. Sharpness in [7] concerns the order of the deficit.

Remark 6.2 (Convexification and the general stability estimates). We retain the established length-preserving convexification reduction [10] in precisely the setting and with the area conventions of [7, Section 3.8]. For simple polygons, A is the usual enclosed area. For self-intersecting polygons, the deduction below is understood within that cited framework: it uses the radial inequality (6.1), preservation of the side lengths, and the nonnegative area increments of the prescribed convexification procedure with a consistent area convention. In particular, it does not assert the same conclusion for an arbitrary alternative definition of area for self-intersecting polygons.

Let $P = P_0, \dots, P_k = P_c$ be the sequence used there, ending at a convex n-gon, and let $T(P) = \sum_{j=0}^{k-1}(A(P_{j+1}) - A(P_j)) \ge 0$ be its total area increase. This is the quantity denoted by $\tau(P)$ in that paper. The operations preserve the multiset of side lengths, so

$$\sigma_s^2(P) = \sigma_s^2(P_c), \qquad \delta_n(P) = \delta_n(P_c) + 4n\tau_n T(P).$$

Consequently the side-length bound (2.4) also holds for P. Applying (6.1) to P itself gives (6.2) with the same C_n, and furthermore

$$T(P) + \sigma_s^2(P) + \sigma_r^2(P) \leq \left(\frac{1}{4n\tau_n} + C_n \right) \delta_n(P).$$

This recovers [7, Theorem 1.1] in the same generality as their convexification reduction, including their non-simple-polygon setting.

For a simple polygon, let $\alpha_c(P) = |P \triangle P_c|$. Each step in the simple-polygon procedure has symmetric-difference area equal to its area increase. The triangle inequality for symmetric differences gives

$$\alpha_c(P) \leq \sum_{j=0}^{k-1} |P_j \triangle P_{j+1}| = T(P).$$

Hence

$$\alpha_c(P) + \sigma_s^2(P) + \sigma_r^2(P) \leq \left(\frac{1}{4n\tau_n} + C_n \right) \delta_n(P),$$

which is [7, Corollary 1.2] with an explicit constant. Together with Corollary 6.1, these deductions give an alternative elementary proof of their main stability estimates, using the classical radial inequality and retaining the same known convexification theorem.

Remark 6.3 (Optimal order of the deficit). The linear order of the deficit in these estimates is optimal. Indeed, take a regular n-gon of circumradius 1 and move one vertex radially outwards by a signed parameter t, keeping the others fixed. For sufficiently small $|t|$ the polygon remains strictly convex. Each of the two adjacent side lengths has derivative $\sin(\pi/n)$ at $t = 0$, while the other side lengths are fixed. It follows that

$$\sigma_s^2(P_t) = \frac{2(n-2)}{n^2} \sin^2 \frac{\pi}{n}\, t^2 + O(t^3).$$

The leading coefficient is positive for every $n \geq 3$. The smooth, nonnegative function $\delta_n(P_t)$ vanishes at $t = 0$, and therefore is $O(t^2)$. The side-length bound gives the reverse estimate, hence $\delta_n(P_t) \asymp t^2$. No estimate with $\delta_n(P_t)^p$ on the right can hold for $p > 1$, with a constant uniform in a neighborhood of the regular polygon. The same argument applies at fixed perimeter after a smooth rescaling: both variances and the deficit are multiplied by the square of the scale factor, which tends to a positive constant.

Indrei's later result [6] also includes angular variance. That additional term is not part of the corollary above.

A targeted literature search located the triangle and quadrilateral cases, the general question in [2], and the stability results cited above, but did not locate the coefficient c_n for general n. No exhaustive bibliographic novelty claim is made.

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Reproducibility

The accompanying source includes a verification script for the algebraic identities, the asymptotic expansion, and numerical checks of the trigonometric inequality, including configurations whose circumcenter is outside the polygon. These computations are supplementary; the proof above is independent of numerical evidence.

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